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SerialProtocolDisplayDefTopic

	
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About Serial Protocol Display

About Serial Protocol Display

Serial Protocol Display is a tool that provides for debug support for high speed serial protocols. This tool debugs "layered" serial protocols (that use 8b/10b encoding) for Protocol Aware instruments. You can use this tool with an executing IG-XL program or as a stand-alone tool. See [Using Serial Protocol Display](#).

	
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UsingSerialProtocolDisplay.2.01



Using Serial Protocol Display

Using Serial Protocol Display

This section describes using Serial Protocol Display to analyze data. The following sections are available:

- [Display Basics](#)
- [Search Control](#)
- [Creating Searches in the Search Panel](#)
- [Properties Panel](#)
- [Capture Setup](#)
- [View and Port Setup Panel](#)
- [Saving and Reading a Transactions File](#)
- [Display Toolbar and Command Reference](#)

To bring up Serial Protocol Display, on the IG-XL Tools tab, in the Launch group, select Tools>Serial Protocol Display.

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UsingSerialProtocolDisplay.2.02



Using Serial Protocol Display : Display Basics

Display Basics

The section covers the basics of viewing and controlling Serial Protocol Display. The following topics are available:

- [Launching Serial Protocol Display](#)
- [Note on Idle sWire Engine](#)
- [Parts of the Display Window](#)
- [Display Grid](#)
- [Notes on Displayed Transactions](#)
- [Display Format](#)
- [Refreshing and Clearing the Display](#)
- [Message Panel](#)
- [Taking and Viewing Snapshots](#)
- [Selecting a Site](#)

- [Grid Font Zoom](#)

See also the following topics that are covered in other sections:

- [Search Control](#)

- [Properties Panel](#)

- [Capture Setup](#)

- [View and Port Setup Panel](#)

- [Saving and Reading a Transactions File](#)

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UsingSerialProtocolDisplay.2.03

	
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Using Serial Protocol Display : [Display Basics](#) : Launching Serial Protocol Display

Launching Serial Protocol Display

You can launch Serial Protocol Display from a running program to capture, view, and save transactions. On the [IG-XL Tools](#) tab, in the Launch group, click Tools and select SerialProtocolDisplay.

You can also launch Serial Protocol Display in stand-alone mode, without IG-XL running: select Start>All Programs>Teradyne IG-XL Vversion>Tools>Serial Protocol Display. See [Saving and Reading a Transactions File](#).

	
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Using Serial Protocol Display : [Display Basics](#) : Note on Idle sWire Engine

Note on Idle sWire Engine

Note that Serial Protocol Display requires the sWire engine to be idle before you can view the transactions.

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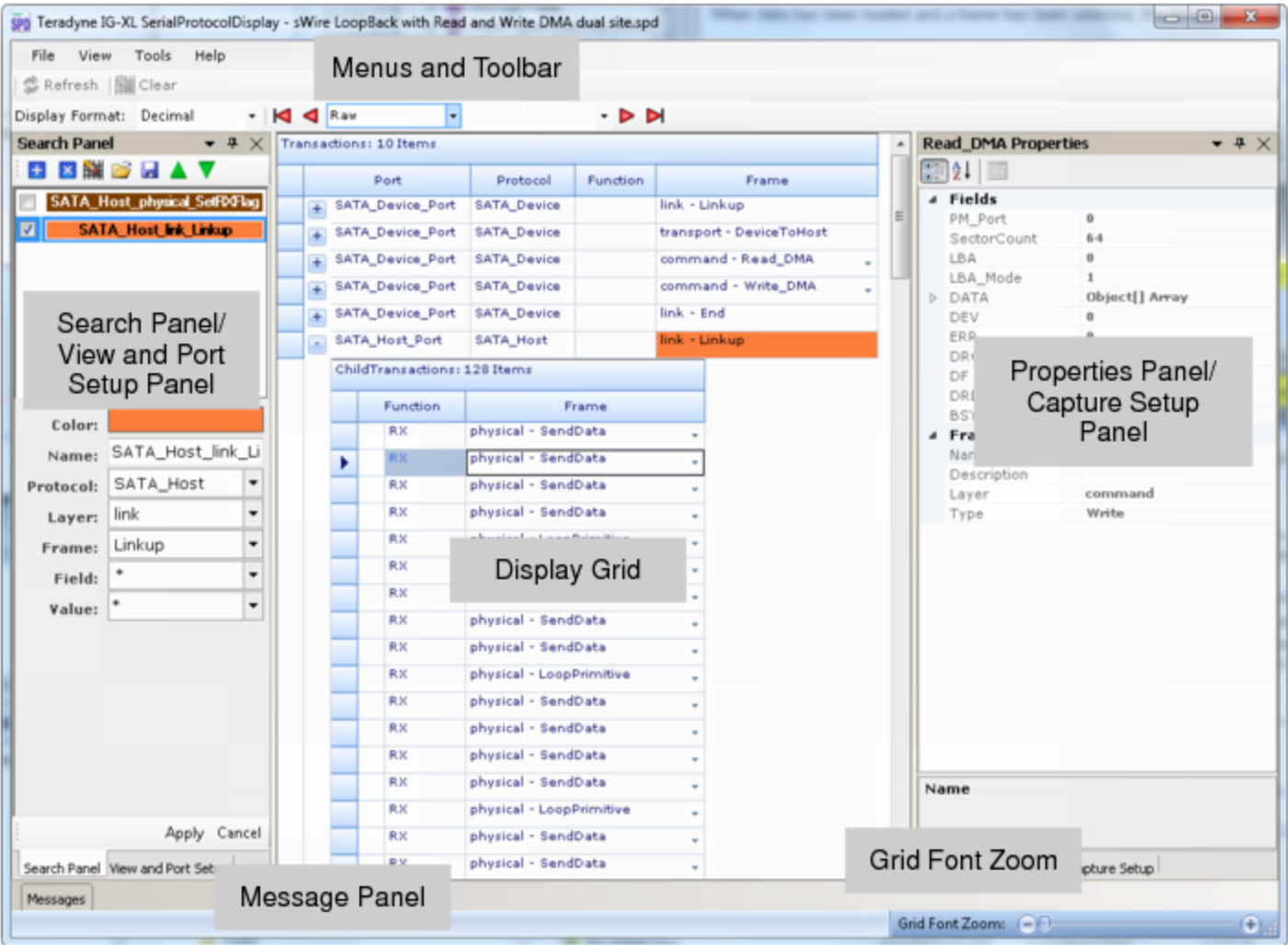
UsingSerialProtocolDisplay.2.05



Using Serial Protocol Display : Display Basics : Parts of the Display Window

Parts of the Display Window

The parts of the Serial Protocol Display window are shown in the following figure.



Serial Protocol Display Window

For information on the parts of the window, see:

•	Display Grid
•	Menus and Toolbar:
–	Display Toolbar and Command Reference
–	Selecting a Site
–	Refreshing and Clearing the Display
–	Display Format
–	Search Control
•	Creating Searches in the Search Panel
•	View and Port Setup Panel
•	Properties Panel
•	Message Panel
•	Grid Font Zoom

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Using Serial Protocol Display : Display Basics : Display Grid

Display Grid

Initially, Serial Protocol Display presents an ordered sequence of captured transactions, one transaction per row:

Transactions: 10 Items				
	Port	Protocol	Function	Frame
+	SATA_Device_Port	SATA_Device		link - Linkup
+	SATA_Device_Port	SATA_Device		transport - DeviceToHost
+	SATA_Device_Port	SATA_Device		command - Read_DMA ▾
+	SATA_Device_Port	SATA_Device		command - Write_DMA ▾
+	SATA_Device_Port	SATA_Device		link - End
+	SATA_Host_Port	SATA_Host		link - Linkup
+	SATA_Host_Port	SATA_Host		transport - DeviceToHost
+	SATA_Host_Port	SATA_Host		command - Read_DMA ▾
+	SATA_Host_Port	SATA_Host		command - Write_DMA ▾
+	SATA_Host_Port	SATA_Host		link - End

Display Grid

You control what ports are displayed by checking and unchecking the boxes in the Ports/Modules panel.

Serial Protocol Display can display data for both transmit and receive protocol links. You can examine packets at all protocol layers, including multiple layers in the grid.

You can drill down to the sublayers of a frame by clicking the plus sign (+) control at the left of each transaction. The layers are displayed below the associated top layer. You can continue to use the + controls on child transactions to expose further layers of child transactions:

Transactions: 10 Items				
	Port	Protocol	Function	Frame
-	SATA_Host_Port	SATA_Host		command - Read_DMA
ChildTransactions: 6 Items				
			Function	Frame
			+	transport - HostToDevice
			-	transport - ReceiveData
ChildTransactions: 1 Items				
			Function	Frame
			-	link - ReceiveFIS
ChildTransactions: 2067 Items				
			Function	Frame
			TX	physical - WaitForPrimitive
			TX	physical - SetRXFlag
			TX	physical - WaitForTXFlag
			RX	physical - SetTXFlag
			RX	physical - WaitForRXFlag
			TX	physical - WaitForPrimitive
			TX	physical - ReceiveData
			TX	physical - ReceiveData
			TX	physical - ReceiveData

Child Transactions

If a transaction has field-level data, you can use the inverted triangle at the right end of the transaction row to display it:

		link - ReceiveFIS
ChildTransactions: 2067 Items		
	Function	Frame
	TX	physical - WaitForPrimitive
	TX	physical - SetRXFlag
	TX	physical - WaitForTXFlag
▶	RX	physical - SetTXFlag
	RX	00000000 K28.3 D21.4 D10.2 D10.2
	TX	physical - WaitForPrimitive

Field-Level Data

You can control the format in which field data is displayed. See Display Format.

Serial Protocol Display highlights failing transactions by coloring a row's leftmost column red:

	Port	Protocol	Function	Frame
▶ +	SATA_Port0	SATA_Device		link - Linkup
+	SATA_Port0	SATA_Device		transport - DeviceToHost
+	SATA_Port0	SATA_Device		command - Write_DMA
+	SATA_Port0	SATA_Device		link - End

Failing Transaction in a Row

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Using Serial Protocol Display : [Display Basics](#) : Notes on Displayed Transactions

Notes on Displayed Transactions

For sWire, only physical layer TX transactions are captured by hardware. The rest of the transactions are reconstructed by Serial Protocol Display based on what was sent.

Serial Protocol Display lets you view encoded or non-encoded data. See [Display Format](#).

The data at the command and transport layers is unscrambled and the data at the link and physical layers is scrambled (assuming scrambling is turned on).

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Using Serial Protocol Display : [Display Basics](#) : Display Format

Display Format

Serial Protocol Display lets you view encoded or non-encoded data. You can control how the field-level data is displayed in the grid and in the Properties Panel. Use this control:

Display Format:

Hex

Display Format Control

The following are the format options

- Name (that is, symbol
- Binary
- Decimal
- Hex
- Binary 8Bit
- Decimal 8Bit
- Hex 8Bit
- ASCII

The screen captures below show how the same field-level data looks in the Properties pane with some different options chosen:

Display Formats

Selected Format

Displayed Values

Name

Fields

ALIGN	Object[] Array
Symbol1	K28.5
Symbol2	D 10.2
Symbol3	D 10.2
Symbol4	D27.3

Hex

Fields

ALIGN	Object[] Array
Symbol1	000000FA
Symbol2	00000155
Symbol3	00000155
Symbol4	00000363

Hex 8Bit

Fields

ALIGN	Object[] Array
Symbol1	(000000FA)
Symbol2	4A
Symbol3	4A
Symbol4	7B

The parentheses around Symbol1 indicate that the value is not in the format you requested. There is no equivalent 8Bit value for "K28.5."

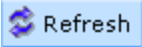
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Using Serial Protocol Display : Display Basics : Refreshing and Clearing the Display

Refreshing and Clearing the Display

Serial Protocol Display must be refreshed when new transactions are captured or when certain events occur that invalidate the current display: for example, executing a test that starts different ports from the previous test.

 When a refresh is needed, Serial Protocol Display shows a message in the window title bar. You can manually refresh the display by clicking this button, using the command View>Refresh, or by pressing F5 or F12.

If the data retrieval is long enough to invoke the progress bar (which is shown in the status bar at the bottom of tool), the Refresh button changes into a Cancel button for the duration of the data load.

 At any point, you can clear all currently displayed transactions from the display and from the tool memory. Use this button, the View>Clear command, or Alt+C.

	
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Using Serial Protocol Display : [Display Basics](#) : Message Panel

Message Panel

Messages that do not require an immediate response are written to the Message Panel. For more information, see [About the Message Panel](#).

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Using Serial Protocol Display : Display Basics : Taking and Viewing Snapshots

Taking and Viewing Snapshots

Serial Protocol Display lets you take snapshots of the current state of the tool. Later, you can view the snapshots for a visual comparison of the tool states.

The [command](#) View>Take Snapshot takes a snapshot of the Serial Protocol Display window. The View>View Snapshot command lets you view the snapshots. It has these options:

Most Recent	Displays the most recent snapshot.
Other	Brings up a browser to let you select the snapshot to view.

The display uses the default Windows image viewer.

By default, a snapshot is saved to the test program folder. If the default folder is read-only, it is saved to the tmp folder. You are not prompted to specify a folder. The snapshot is saved as JPEG file with a filename of this form:

SerialProtocolDisplay_YYMMDDHHMMSS.jpg

This means that snapshots in the browser are in chronological order.

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Using Serial Protocol Display : Display Basics : Selecting a Site

Selecting a Site

Serial Protocol Display shows the data for one site at a time. You can use the site selector to specify a site:

Site:

1

 Master

Serial Protocol Display Site Selection

If you click Master, Serial Protocol Display uses the master site as it is defined in DataTool. The site number displays the current master site, but it is grayed out because you cannot change it.

If you disable Master, you can manually select the site to be displayed.

If you have enabled Master and the master site changes in IG-XL, Serial Protocol Display indicates that you must refresh the display.

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Using Serial Protocol Display : [Display Basics](#) : Grid Font Zoom

Grid Font Zoom

The bottom of the Serial Protocol Display window has a sliding zoom control:



Grid Font Zoom Control

The control increases or decreases the size of the font in the display grid. As you move the slider, a box appears above the control showing the font size as specified by the current position.



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Using Serial Protocol Display.2.14

	
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Using Serial Protocol Display : Search Control

Search Control

The search control lets you go to first, previous, next, or last instance of the specified item in the display. The items are Fail, Defined, and Raw.

The Fail option takes you to the first, previous, next, or last failed transaction in the display:



Search for Fail

The Defined option executes one of the searches that you have defined in the Search panel. If you select Defined, an additional combo box appears containing the names of the searches currently defined in the Search panel:



Defined Search

For more information, see [Creating Searches in the Search Panel](#).

The Raw option lets you select the search criteria directly in the search control. As you select the criterion for one level from the drop-down list, a combo box appears containing the valid values for the next level down, and so forth, until you have a complete search:

Raw

SATA_Host

link

ReceiveFIS

Transactions: 10 Items

	Port	Protocol	Function	Frame
+	SATA_Device_Port	SATA_Device		link - Linkup
+	SATA_Device_Port	SATA_Device		transport - DeviceToHost
+	SATA_Device_Port	SATA_Device		command - Read_DMA
+	SATA_Device_Port	SATA_Device		command - Write_DMA
+	SATA_Device_Port	SATA_Device		link - End
+	SATA_Host_Port	SATA_Host		link - Linkup
-	SATA_Host_Port	SATA_Host		transport - DeviceToHost

ChildTransactions: 1 Items

	Function	Frame
▶ +		link - ReceiveFIS

Raw Search

For any level, you can select an asterisk (*), which matches all values.

Note that any of these searches applies only to the currently checked ports. Even if the search criteria match an item in a port that is not checked, the search does not find the item.

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Using Serial Protocol Display : Creating Searches in the Search Panel

Creating Searches in the Search Panel

The Search Panel lets you create and save sets of search criteria. You can also write the searches to a file, which you can reload later.

The following topics are available:

- [Adding and Editing a New Search](#)
- [Using Colors in Search](#)
- [Removing and Saving Searches](#)

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Using Serial Protocol Display.2.16



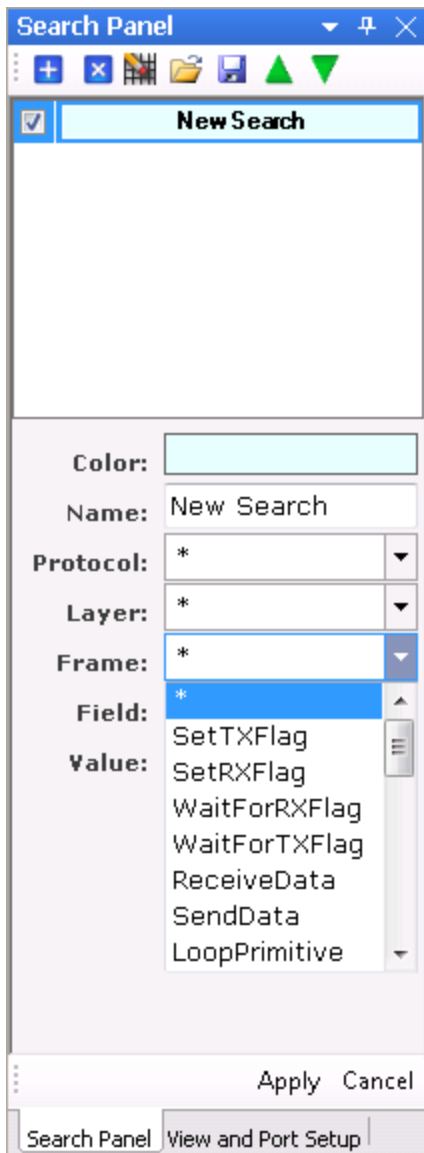
Using Serial Protocol Display : [Creating Searches in the Search Panel](#) : Adding and Editing a New Search

Adding and Editing a New Search

To add a new search, you can:

- [Click the](#) Add button on the Search Panel
- [Right-click the desired target in the display grid](#)

 If you click the Add control at the top of the Search Panel, a new, empty search is added to the panel:



New Search

As the figure shows, each search element has a drop-down list containing valid values for that element. You can select the value for each element. By default, the element has an asterisk (*), meaning that it matches all values.

Another way to add a search is to find the target you want in the display grid. If you right-click the target, the menu displays that item, plus all fields that the item contains:

Transactions: 10 Items

	Port	Protocol	Function	Frame
+	SATA_Device_Port	SATA_Device		link - Linkup
+	SATA_Device_Port	SATA_Device		transport - DeviceToHost
+	SATA_Device_Port	SATA_Device		command - Read_DMA
▶	SATA_Device_Port	SATA_Device		command - Write_DMA
+	SATA_Device_Port	SATA_Device		link - End
+	SATA_Host_Port	SATA_Host		link - Linkup
+	SATA_Host_Port	SATA_Host		transport - DeviceToHost
+	SATA_Host_Port	SATA_Host		command - Read_DMA
+	SATA_Host_Port	SATA_Host		command - Write_DMA
+	SATA_Host_Port	SATA_Host		link - End

New search - Write_DMA

New search - PM_Port (0)

New search - SectorCount (64)

New search - LBA (0)

New search - LBA_Mode (1)

New search - DATA (22676545...)

New search - DEV (0)

New search - ERR (0)

New search - DRQ (0)

New search - DF (0)

New search - DRDY (1)

New search - BSY (0)

Search Target Context Menu

When you select an item from the menu (in this example, "DEV"), the tool fills in the elements in the Search Panel:

Search Panel

☒ SATA_Device_command_Write_DMA

Color:

Name: SATA_Device_command_Write_DMA

Protocol: SATA_Device

Layer: command

Frame: Write_DMA

Field: DEV

Value: 0

Search Filled from Grid

Once you add a new search, you can edit it. You can directly edit the Name element. The other elements have drop-down lists with the valid values, plus the asterisk (*) for matching all values for

the element.

On changing the color, see [Using Colors in Search](#).

To undo a change, including a color change, click [Cancel](#) at the bottom of the Search Panel.

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Using Serial Protocol Display : Creating Searches in the Search Panel : Using Colors in Search

Using Colors in Search

You can assign each search a different color, so that the matched results for each search can be differentiated in the display grid:

Search Panel

☒ Host_link_ReceiveFIS
☒ Dev_cmd_Write_DMA
☒ Dev_transport_ReceiveData

Color:

Name:

Protocol:

Layer:

Frame:

Field:

Value:

Apply Cancel

Transactions: 10 Items				
	Port	Protocol	Function	Frame
+ []	SATA_Device_Port	SATA_Device		transport - DeviceToHost
+ []	SATA_Device_Port	SATA_Device		command - Read_DMA
- []	SATA_Device_Port	SATA_Device		command - Write_DMA
ChildTransactions: 10 Items				
			Function	Frame
[+]			transport - HostToDevice	
[+]			transport - DMA_Activate	
[+]			transport - ReceiveData	
[+]			transport - DMA_Activate	
[+]			transport - ReceiveData	
[+]			transport - DMA_Activate	
[▶]			transport - ReceiveData	
[+]			transport - DMA_Activate	
[+]			transport - ReceiveData	
[+]			transport - DeviceToHost	
+ []	SATA_Device_Port	SATA_Device		link - End
+ []	SATA_Host_Port	SATA_Host		link - Linkup
- []	SATA_Host_Port	SATA_Host		transport - DeviceToHost
ChildTransactions: 1 Items				
			Function	Frame
[+]			link - ReceiveFIS	
+ []	SATA_Host_Port	SATA_Host		command - Read_DMA
+ []	SATA_Host_Port	SATA_Host		command - Write_DMA

Use of Color in Searches

By default, all searches use the same color. To change a color, double-click the Color box. A Color window comes up, letting you choose a color.

If two searches match the same display item, the color displayed is that of the higher search in the Search Panel list. For example, the two searches shown in the following figure have the same criteria. However, the first one uses asterisks for Field and Value, while the second has actual values.

The color of the first search has precedence in the display:

Search Panel

Color:

Name: RecDataAsterisks

Protocol: SATA_Device

Layer: transport

Frame: ReceiveData

Field: *

Value: *

Transactions: 10 Items

	Port	Protocol	Function	
+	SATA_Device_Port	SATA_Device		link - Linkup
+	SATA_Device_Port	SATA_Device		transport - Devi
+	SATA_Device_Port	SATA_Device		command - Rea
-	SATA_Device_Port	SATA_Device		command - Wri

ChildTransactions: 10 Items

	Function	Frame
+		transport - HostToDevice
+		transport - DMA_Activate
▶		transport - ReceiveData
+		transport - DMA_Activate
+		transport - ReceiveData
+		transport - DMA_Activate
+		transport - ReceiveData
+		transport - DMA_Activate
-		transport - ReceiveData

Color Affected by Order of Searches

▲ ▼ To have the other color displayed, you must change the order of the searches, using the Up and Down buttons in the Search Panel controls. When you reverse the order of searches, the search with values for the elements has precedence, and its color is used in the display, as shown in the following figure.

Search Panel

☒ **RecDataValues**

☒ **RecDataAsterisks**

Transactions: 10 Items

	Port	Protocol	Function	
+	SATA_Device_Port	SATA_Device		link - Linku
+	SATA_Device_Port	SATA_Device		transport -
+	SATA_Device_Port	SATA_Device		command -
-	SATA_Device_Port	SATA_Device		command -

ChildTransactions: 10 Items

	Function	Frame
+		transport - HostToDevice
+		transport - DMA_Activate
▶		transport - ReceiveData
+		transport - DMA_Activate
+		transport - ReceiveData
+		transport - DMA_Activate
+		transport - ReceiveData
+		transport - DMA_Activate
-		transport - ReceiveData

Color:

Name:

Protocol:

Layer:

Frame:

Field:

Value:

Transaction Details

Port	Protocol	Function	Frame	Field	Value
SATA_Device_Port	SATA_Device		transport - HostToDevice	PM_Port	(00000000)
SATA_Device_Port	SATA_Device		transport - DMA_Activate	PM_Port	(00000000)
SATA_Device_Port	SATA_Device		transport - ReceiveData	PM_Port	(00000000)
SATA_Device_Port	SATA_Device		transport - DMA_Activate	PM_Port	(00000000)
SATA_Device_Port	SATA_Device		transport - ReceiveData	PM_Port	(00000000)
SATA_Device_Port	SATA_Device		transport - DMA_Activate	PM_Port	(00000000)
SATA_Device_Port	SATA_Device		transport - ReceiveData	PM_Port	(00000000)
SATA_Device_Port	SATA_Device		transport - DMA_Activate	PM_Port	(00000000)
SATA_Device_Port	SATA_Device		transport - ReceiveData	PM_Port	(00000000)
SATA_Device_Port	SATA_Device		transport - DMA_Activate	PM_Port	(00000000)

Search Order Reversed

	
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Using Serial Protocol Display : Creating Searches in the Search Panel : Removing and Saving Searches

Removing and Saving Searches

To remove the effects of a search from the display grid without deleting the search from the Search Panel list, uncheck the box next to the search in the list:

☒	RecDataValues
☐	RecDataAsterisks
☒	SATA_Device_physical WaitForT

Unchecking Searches

Checking the box again reapplies the search.

 To remove a search from the list, select the search and then click the Clear button at the top of the panel.

 To remove all searches, click the Clear All button.

 To save searches to a file, click the Save button. A browser lets you specify the name for the file. A file with the extension .xml is created. This saves all the current searches, along with the state of the check box.

 To load a file of previously saved searches, click the Open button. A browser lets you navigate to the file.

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Using Serial Protocol Display : Properties Panel

Properties Panel

The Properties panel displays the properties of the frame that is currently selected in the display grid or of the port currently selected in the Ports/Modules panel.

If the Properties window is not visible, use View>Properties to display it.

The Properties window can be divided into categories. You can specify how properties are to be displayed using the buttons at the top of the Properties window:

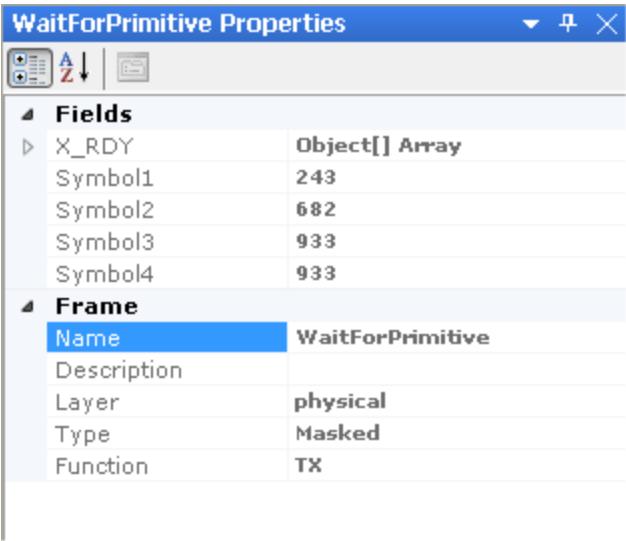


Categorized, for example, Fields, Frame.



Alphabetical, without category dividers.

The properties listed depend on the frame or port currently selected in the display grid. The following figure shows the properties for a frame that has fields:



Properties Panel Categorized by Fields

If a field is an array, you can click the triangle control to display the individual items:

ReceiveFIS Properties

Fields

DATA

Object[] Array

[0]

-1038551382

[1]

-14240920

[2]

-1526185108

[3]

877843220

[4]

-1974102782

CRC

1675180359

Frame

Name

ReceiveFIS

Description

Layer

link

Type

Read

Displaying an Array

The Display Format control affects the display in the Properties Panel as well as in the display grid.

See Display Format.

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Using Serial Protocol Display : Capture Setup

Capture Setup

Part of the interactive debug process includes setting up HRAM triggering to capture transactions of interest. You do this using the Capture Setup panel. The panel displays the Capture trigger setup properties for the port selected.

Capture properties are set to capture the maximum number of transactions by default.

To enable the setup control, check the box Use custom capture setup. Then, from the Select Port drop-down, select the port for which you want to set up capture:

Capture Setup

☒ Use custom capture setup

Select Port:

SATA_Device_Port

▲ - sWire HRAM

Trigger Type:

Fail

☐ Wait for Event

Capture Events

☐ DisparityError

☐ CodeError

☐ Timeout

☐ ReceiveAlignTrigger

Timeout Value:

Trigger Mode:

HRAM Status

☐ Armed

☐ Triggered

☐ Overflowed

Read HW

SATA_Port Pro...

Capture Setup

Capture Setup Panel

For more information, see [Help](#) for the specific instrument.

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file:///C:/Users/cpatle2/AppData/Local/Temp/chm_extract_57q4l6uf/combined.html

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UsingSerialProtocolDisplay.2.21



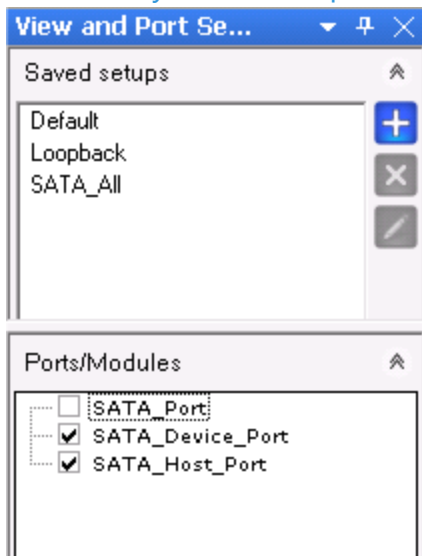
Using Serial Protocol Display : View and Port Setup Panel

View and Port Setup Panel

The Port/Modules panel determines which ports appear in the display grid. You can use the View and Port Setup panel to create view setups. A view setup lets you select ports that you want included in the display and save the selection as a named setup. You can later view the same display by double-clicking the name of the setup.

Available setups are listed in the View and Port Setup panel. If this is not visible, use View>View and Port Setup to make it visible.

The following shows one built-in setup and two user-created setups. The currently selected setup causes only two of the ports to be displayed:



View and Port Setup Panel

The default built-in setup is Default, which is equivalent to checking all boxes in the Ports/Modules panel.



To create a view setup, select the items that you want included by checking their boxes in the Ports/Modules window. Then click the Add button. A new string is added to the Saved setups

window with the default name NewViewN. The name is in edit mode, so you can change it immediately.

Once you add a setup, you can double-click the name to display the view.



To delete a setup, select it in the Saved setups window and click the Delete button. To delete all user-created setups, right-click in the white area of the Saved setups window (not on a setup name) and select Delete All from the context menu.



To modify a setup, select it in the Saved setups window. Make changes to the check boxes. Then click the Update button.

To rename a setup, select it in the Saved setups window, right-click, and select Rename. Note that Apply, Delete, and Update are also on the right-click context menu.



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Using Serial Protocol Display : Saving and Reading a Transactions File

Saving and Reading a Transactions File

Serial Protocol Display lets you debug transaction specifics offline. You can save captured transactions to a file and then open the file later to continue debugging the transactions without the tester. You can open the file while connected to IG-XL or in stand-alone mode.

Serial Protocol Display translates the transactions to XML format and saves them to a file with the .spd extension.

To save the currently captured transactions, use File>Save (Alt+S) or Save As. The first time you save to a file, the browser comes up to let you specify a filename and location.

Note that only transactions for the currently specified site are saved. See [Selecting a Site](#).

To load a saved file, use File>Open (Alt+O). A browser lets you browse to the saved .spd file.

When you open a file, you are disconnected from IG-XL. This is the same state as bringing up the tool in stand-alone mode and loading a saved file.

In this disconnected state, any Serial Protocol Display functions that interact with IG-XL or that deal with capturing transactions are disabled. The main functions that are still enabled are:

- Changing the display format. See [Display Format](#).
- Using the Search control and the Search Panel. See [Search Control](#) and [Creating Searches in the Search Panel](#).
- Using the View and Port Setup panel to specify which ports to display and to create view setups. See [View and Port Setup Panel](#).

You can reconnect to IG-XL using the command Tools>Connect to IG-XL. Connecting to IG-XL clears the data displayed from the file.

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UsingSerialProtocolDisplay.2.23



Using Serial Protocol Display : Display Toolbar and Command Reference

Display Toolbar and Command Reference

This section lists the actions that are available from the Serial Protocol Display menus and toolbars. Links to more complete descriptions are included.

Most toolbar buttons correspond to menu commands, but some are available only on the toolbars, on the menu, or on the right-click context menus.

The section is organized on where the command appears, on the toolbar or on one of the menus:

- | | |
|---|-------------------------------------|
| • | Toolbar |
| • | File Menu Commands |
| • | View Menu Commands |
| • | Tools Menu Commands |



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Using Serial Protocol Display : Display Toolbar and Command Reference : Toolbar

Toolbar

The Serial Protocol Display toolbar looks like this:



Serial Protocol Display Toolbar

The individual controls are as follows:

	Site control Selects which site to display: either the master site, specified in DataTool, or a site from the drop-down list. See Selecting a Site .
	View>Refresh (F5, F12) Refreshes the data. The tool prompts for when a refresh is necessary. See Refreshing and Clearing the Display .
	View>Clear (Alt+C) Clears all items from the display grid. See Refreshing and Clearing the Display .
	Display Format Specifies how numeric data is displayed in the field-level displays and in the Properties panel. See Display Format .
	Search

Searches for a fail, for criteria you specify here, or for a defined set of criteria. See [Search Control](#).

		
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Using Serial Protocol Display : Display Toolbar and Command Reference : File Menu Commands

File Menu Commands

The File menu has commands for opening and saving files. See [Saving and Reading a Transactions File](#).

Open	Alt+O Opens an existing Serial Protocol Display capture file (extension .spd).
Save	Alt+S Saves the displayed data to the file that you have opened. If this is a new file that has not been saved before, it performs a Save As.
Save As	Saves the displayed data to a new file that you specify.
Exit	Alt+X Exits from the tool.



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Using Serial Protocol Display : Display Toolbar and Command Reference : View Menu Commands

View Menu Commands

View menu commands let you zoom in and out on the Frame window:

Messages	Displays the Messages panel and gives it focus. See About the Message Panel.
View and Port Setup	Shows the View and Port Setup panel. See View and Port Setup Panel.
Capture Setup	Shows the Capture Setup panel. See Capture Setup.
Properties	Shows the Properties panel. See Properties Panel
Search Panel	Shows the Search Panel. See Search Control.
Take Snapshot	Takes a snapshot of the display section. See Taking and Viewing Snapshots.
View Snapshot	Displays the snapshots. A pull-right menu lets you view the most recent snapshot or select a different snapshot. See Taking and Viewing Snapshots.
Clear	 Clear Alt+C Clears all items from the display grid.
Refresh	 Refresh F5, F12 Refreshes the display to reflect the latest state of the system. See Refreshing and Rebursting.

Restore Design View

Restores the tool to its initial default condition. For example, panels that were dismissed are redisplayed, the grid is turned on, cursors are turned off, and initial row heights are restored.

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Using Serial Protocol Display : Display Toolbar and Command Reference : Tools Menu Commands

Tools Menu Commands

These commands bring up other digital tools or connect to IG-XL.

Waveform Display	Brings up the Waveform Display tool. See About Waveform Display .
Protocol Studio	Opens Protocol Studio to display the transactions executed during this session. See About Protocol Studio .
Connect to IG-XL	Reconnects to the IG-XL workbook from stand-alone mode or after disconnecting by loading a saved file. See Saving and Reading a Transactions File .



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